

Closed Forms for Recursive Successive-Differences Problems

A recursive successive-differences problem can often be converted into a closed form by treating the differences as a sequence of their own. Once a formula for the differences is known, summing those differences recovers the original sequence.

1 First Differences

Suppose a sequence $a_0, a_1, a_2, \dots$ satisfies

$$a_n - a_{n-1} = d_n, \quad n \geq 1,$$

where d_n is the sequence of first differences. Summing both sides from $k = 1$ to $k = n$ gives

$$\begin{aligned} \sum_{k=1}^n (a_k - a_{k-1}) &= \sum_{k=1}^n d_k, \\ a_n - a_0 &= \sum_{k=1}^n d_k. \end{aligned}$$

The sum on the left telescopes, so

$$a_n = a_0 + \sum_{k=1}^n d_k.$$

Thus the main task is to find a formula for d_n and then evaluate its sum.

For example, if the first differences are constant,

$$a_n - a_{n-1} = c,$$

then $d_n = c$, and therefore

$$a_n = a_0 + cn.$$

This is the closed form of an arithmetic sequence.

2 Second Differences

Now suppose the second differences are constant:

$$(a_n - a_{n-1}) - (a_{n-1} - a_{n-2}) = c, \quad n \geq 2.$$

Equivalently,

$$a_n - 2a_{n-1} + a_{n-2} = c.$$

Define the first-difference sequence by

$$d_n = a_n - a_{n-1}.$$

The recurrence for the second differences becomes

$$d_n - d_{n-1} = c.$$

Therefore d_n is an arithmetic sequence:

$$d_n = d_1 + (n-1)c.$$

Using the first-difference formula from the previous section,

$$\begin{aligned} a_n &= a_0 + \sum_{k=1}^n d_k \\ &= a_0 + \sum_{k=1}^n (d_1 + (k-1)c) \\ &= a_0 + nd_1 + \frac{cn(n-1)}{2}. \end{aligned}$$

Since $d_1 = a_1 - a_0$, the closed form may be written as

$$a_n = a_0 + n(a_1 - a_0) + \frac{cn(n-1)}{2}.$$

The right-hand side is a quadratic polynomial in n . In general, a sequence with constant second differences is quadratic.

3 The General Finite-Difference Rule

The forward-difference operator is defined by

$$\Delta a_n = a_{n+1} - a_n.$$

Repeated differences are defined recursively:

$$\Delta^2 a_n = \Delta(\Delta a_n), \quad \Delta^3 a_n = \Delta(\Delta^2 a_n),$$

and similarly for higher orders.

If the r -th differences are constant, then a_n is a polynomial in n of degree at most r . Its closed form can be written in the Newton forward-difference form

$$a_n = a_0 + \binom{n}{1} \Delta a_0 + \binom{n}{2} \Delta^2 a_0 + \cdots + \binom{n}{r} \Delta^r a_0.$$

Equivalently, if $\Delta^0 a_0$ is understood to mean a_0 , then

$$a_n = \sum_{j=0}^r \binom{n}{j} \Delta^j a_0.$$

Here

$$\binom{n}{j} = \frac{n!}{j!(n-j)!}$$

is a binomial coefficient. The values

$$a_0, \quad \Delta a_0, \quad \Delta^2 a_0, \quad \dots, \quad \Delta^r a_0$$

are the first entries in the rows of a finite-difference table.

4 Worked Example

Consider the sequence

$$2, 5, 10, 17, 26, \dots$$

and index its first term as $a_0 = 2$. Its first differences are

$$3, 5, 7, 9, \dots,$$

and its second differences are

$$2, 2, 2, \dots$$

Thus

$$a_0 = 2, \quad \Delta a_0 = 3, \quad \Delta^2 a_0 = 2.$$

The general finite-difference formula gives

$$\begin{aligned} a_n &= a_0 + \binom{n}{1} \Delta a_0 + \binom{n}{2} \Delta^2 a_0 \\ &= 2 + 3n + 2 \binom{n}{2} \\ &= 2 + 3n + n(n-1) \\ &= n^2 + 2n + 2. \end{aligned}$$

Therefore, with indexing beginning at $n = 0$,

$$\boxed{a_n = n^2 + 2n + 2}.$$

For instance, substituting $n = 0, 1, 2$ produces

$$a_0 = 2, \quad a_1 = 5, \quad a_2 = 10,$$

as required.

If the same sequence is instead indexed beginning with $a_1 = 2$, then its closed form is

$$\boxed{a_n = n^2 + 1}.$$

This difference illustrates why the starting index must be identified before writing a closed form.